

Detection Segmentation and Noise Removal Process of Cancerous Image Cell of Liver Through Modified Marker-Controlled Watershed Approach



Tahamina Yesmin

Abstract We know that CT scan, MRI and X-rays scan processes are there in medical image analysis to identifying the disease but from them also sometime we don't get the clear and detailed information. Various studies have been presented and developed methods. Not effecting the idea of previous studies this study is going to represent a new way of seeing the segmentation approach through improved marker-controlled approach with the help of Sobel edge detection method to develop a new model of analysis the medical image so that we can get the less segmented image to reduce the noise coming from unprocessed image and get a clear view of bio-medical image. The process would help in early detection of cancer cells and helps in spreading it to other part with the help of proper diagnosis. This study also proposed the statistical analysis of images of which will receive after the proposed algorithm applied.

Keywords Morphology · Marker-controlled · Cancer · Segmentation · Medical · Image

1 Introduction

The liver is the largest solid organ in the human body and plays an essential role in metabolism and digestion. Worldwide, primary liver cancer is the second most common fatal cancer. It is assessed that in excess of 3000 individuals were determined to have liver cancer in 2023. The typical age at finding is 69 years of age. Therefore, it always requires to improve the imaging techniques always where as we can get a clearer image and improve the quality of image as much as possible with the identification and description of components while imaging. Medical images give specialists rich assistant data to assist them with making decisions. As a significant piece of medical picture handling, picture division is of incredible importance for

T. Yesmin (✉)
Adamas University, Kolkata, WB, India
e-mail: ytahamina@gmail.com

precise ID, itemized investigation, sensible finding, expectation and avoidance of cancer. While work on segmentation approach, we may need to work on image shape, size, grayscale, boundaries which need to improve or clear, complex texture changes position changes, extract local values etc. so the selection of algorithms and their pre-processing part is must play important role in segmentation. In this paper we are going to introduce marker-controlled watershed approach with Sobel edge detection approach. The use of marker-based approach helps in reducing noise and help to improve the image segmentation part to make the image clearer [1–3].

1.1 Causes Behind Liver Cancer

Basically, liver cancer harmful tumor that start unknowingly grow in the liver. It may form in different ways like:

HCC (Hepatocellular Carcinoma) which also known as hepatoma is most common liver cancer that start effecting the main cell type (hepatocytes) present in liver. Another type of cancer is bile duct cancer (cholangiocarcinoma) which start effecting in the cells lining that connected with the bowel and gall bladder and the other one which is angiosarcoma that start effecting in the blood vessels, this type of cancer is very rare type which occurs mainly in the aged group of people. The other type of causes behind liver cancer are fatty liver with various grade, jaundice due to hepatitis B and C and the mutation of DNA especially for Hepatitis C and many secondary causes as well. Hepatitis infections are key entertainers in the advancement of hepatocellular carcinoma (HCC), one of the main sources of death in disease [4].

Hepatitis B or C infections which bring about long haul (ongoing) disease can cause liver malignant growth. This is the greatest realized risk factor for essential liver disease in Australia with the hepatitis infection.

The secondary causes of cancer in the liver which start occurring in another part of body but somehow spread to the liver for example bowel cancer with liver or sometime cells break away from the primary cancer cell, these cancer cells carried in the bloodstream to another part of the body. If cancer started in anyone's lung and start spreading to liver therefore the areas of cancer in the liver are made up of cancer cells [5].

2 Literature Reviews [6–10]

Author(s)	Method/ Architecture	Dataset	Key findings	Research gaps
Sabir, M.W., et al	ResU-Net (Dense-Net + U-Net)	3D-IRCADb01	Effective segmentation of liver tumors; strong performance with pre-processing and data augmentation	Lack of noise removal methods; noise reduction could enhance segmentation
Hansch, A., et al	3D U-Net with multi-model training	MRI/CT data	High accuracy in liver segmentation; better performance than traditional models	Need for a more representative validation set; risk of overfitting not addressed
Biao, W. & Chunfeng, Y	Review of U-Net-based CNNs (2D, 2.5D, 3D)	N/A	Summarizes benefits and issues of U-Net architectures; highlights necessity of deep learning	Focuses on classification of CNNs, not on noise reduction or over-segmentation
Wang, J., et al	UNet + +	MR images	Automatic segmentation of liver tissues; evaluates using DSC and AFPR metrics	Methodology focuses on detection, lacks noise consideration
Jha, D., et al	PVTFormer with advanced residual up sampling	CT images	Improved segmentation masks through hierarchical decoding; effective for healthy liver segmentation	Noise removal methods not included; could further enhance results

2.1 Marker-Controlled Watershed Segmentation with Sobel Edge Detector

MCWS is basically a powerful algorithm which generally used in image processing and computer vision areas with some basic steps towards improvements of the algorithm. In this paper the steps followed by sobel and gradient specific. First of all, we need to specify a function which can identify the objects of interest where the dark regions can identify through objects for segmentation. Generally, one marker represents one object so it's very important to extract the marker properly otherwise we may lead into poor segmentation process. After that the main task of the algorithm start with calculating foreground and background marker. The foreground markers connected with pixels within the objects and placed at centre of the objects which helps in the segmentation process and the background marks of pixel which are not the part of any objects they only placed in the areas covering the objects for segmentation

process. If the function modification is needed to calculate the minima of foreground and background markers then we can do it through a constraint on modification function to confirm the markers presence in segmentation process. After all the process we can now work on watershed transformation phase of marker-controlled process to segment the objects properly and perfectly through separate the regions with the help of catchment basins that developed by markers. The marker process application can be done through many different types of pre-processing phases depend on the algorithm which leads to reduce the segmentation process and give the image clarity in a developed way. The markers process now a days also applying to improve the video analysis processes as well. It more helpful for the researchers to work on the segmentation of complex structure image and more difficult observational images [11–14].

Edge mainly helps to gain meaningful information from the image by acting as a boundary between an object and background. From the proper boundary's information, we can segment an image in more meaningful way. Edge can define as borderline pixels which can built a connection between two mutually exclusive regions which differentiate with the luminance values like amplitude, orientation and location of a region. Various types of edge detection method presents and among them the sobel edge detection is much more appropriate for MRI image type, we came to know from the analysis. Here in our experiment, we have taken MRI images [15, 16].

A surmised magnitude of the gradients is estimated by the number of outright values of the slope in the two directions. Sobel edge detector gauge strategy to work out the worth and direction of the inclination by subbing complex cycles by equipment cost investment funds, customary non-most extreme concealment versatile thresholding block association, and traditional hysteresis thresholding. The approach utilizes a couple of 3×3 matrix of complexity masks, in different directions. It is unquestionably delicate to noise in pictures, it really features them as edges. For that reason, this administrator is suggested in enormous data correspondence found in data move. It likewise used to find the estimated outright inclination magnitude for every inputted gar-scale picture [17] (Figs. 1 and 2).

Fig. 1 Original color image of liver tumor

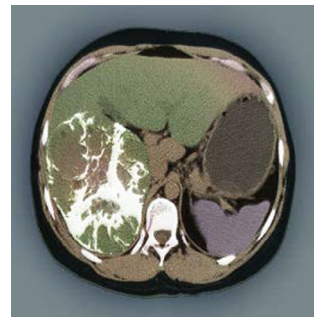
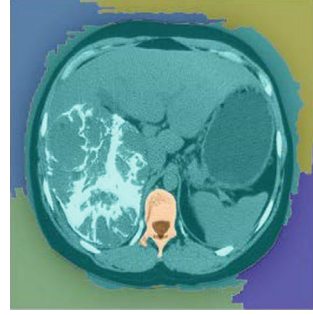


Fig. 2 Marker-controlled image with sobel



2.2 Morphological Reconstruction and Filtering Process Through Mathematical Expression

The mathematical expressions and morphological representation will be executed through erosion, dilation, opening and closing. The morphology through mathematical expression tools and expressions mainly used to extract the needful and useful components for image representation and description. The set-theoretic technique uses for mathematical representation with a quantitative data which will provide object boundaries, skeletons, edge detection for pre and post processing of an image. In this paper for the marker-controlled watershed process we mainly used dilation and erosion transformation.

Erosion:

For an image X and structuring element Y in the A^2 , so it will be expressed through the following equation:

$$X - Y = \{A \in E | Y \approx \in X\} \quad (1)$$

The above equation explaining about the process of erosion of X by Y is actually set of all points A so that Y can shifted by A , is a subset of X i.e., Y is fully constant with X . Y is totally contained through X . To remove the unwanted or unused pixels from the boundary of object we mainly use the erosion process.

Dilation:

The process of dilation can be expressed through following equation:

$$X \oplus Y = \{A \in E | (Ys) \approx nX \neq \emptyset\} \quad (2)$$

The above equation explaining about the process of dilation of X with Y is a set of all replaces, A , so that (Y) and X overlap by minimum of one element. The dilation process actually adds the pixel to the object boundaries of an image. The pixels depend on the structuring elements shape size.

In this paper the concept of 'opening' and 'closing' also used after the dilation and erosion.

Opening of an image is mainly the erosion process followed by dilation which explain by following equation:

$$X \circ Y = \{\text{OPEN}(X, Y) = (X \ominus Y) \oplus Y\} \quad (3)$$

And closing of an image is mainly the dilation process followed by erosion explain by following equation:

$$X \circ Y = \{\text{CLOSE}(X, Y) = (X \oplus Y) \ominus\} \quad (4)$$

2.3 Dataset

To propel clinical image segmentation strategies and to give an objective examination of arising liver cancer division techniques, The LiTS17 dataset distributed by MICCAI. The images also being work and modified for the experimental process. The LiTS17 dataset is the authority dataset published by the opposition to the increasingly live growth division, mutually coordinated by two associations, MICCAI and ISBI [18, 19].

3 Methodology

The proposed algorithm has started working by reading the original MRI images. After reading the original images then the images converted into grayscale to reduce the image segmentation complexity issues to compute in a more simplified way. Then we calculated the image complement along with the image gradient magnitude to bring out the essential information from image within a calculated range which focus on objects and their regions to help in the next phase of algorithm. After that in the next phase of algorithm the process of marker will start with the help of proper edge detector which is Sobel. The markers are mainly use to mark the regions as these are the main areas from where we will get and work on image segmentation for image quality and noise erosion process. The marker will play work by mainly focusing the foreground and background calculation. From here we can mark the needed portions of objects which will helps in proper segmentation. After that the Sobel edge detector will be applied to finish the segmentation process and we will get the final segmented image (Figs. 3 and 4).

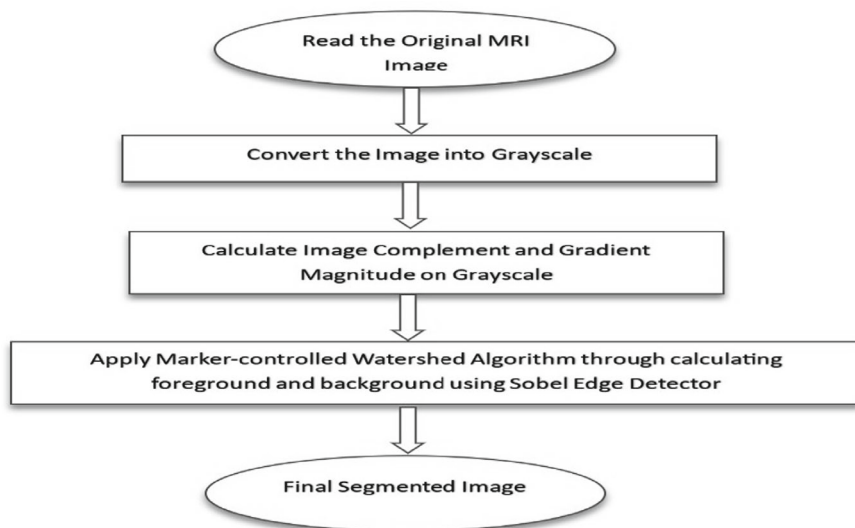


Fig. 3 Methodology of proposed algorithm

4 Results and Discussion

The outcomes of proposed algorithm have shown in the above figures. In Fig. 5 the original MRI images has been converted into gray scale image and then the image complement and image calculate the gradient of that image to work for marker-controlled watershed approach with the sobel edge detection process, where we can see the improvement in segmentation approach through the statistical measurements that will be state in following which also offer the early disease detection process through the laboratory diagnosis process. The proposed algorithm has worked properly as we can extract the marked part and see the noise have also reduced (Table 1).

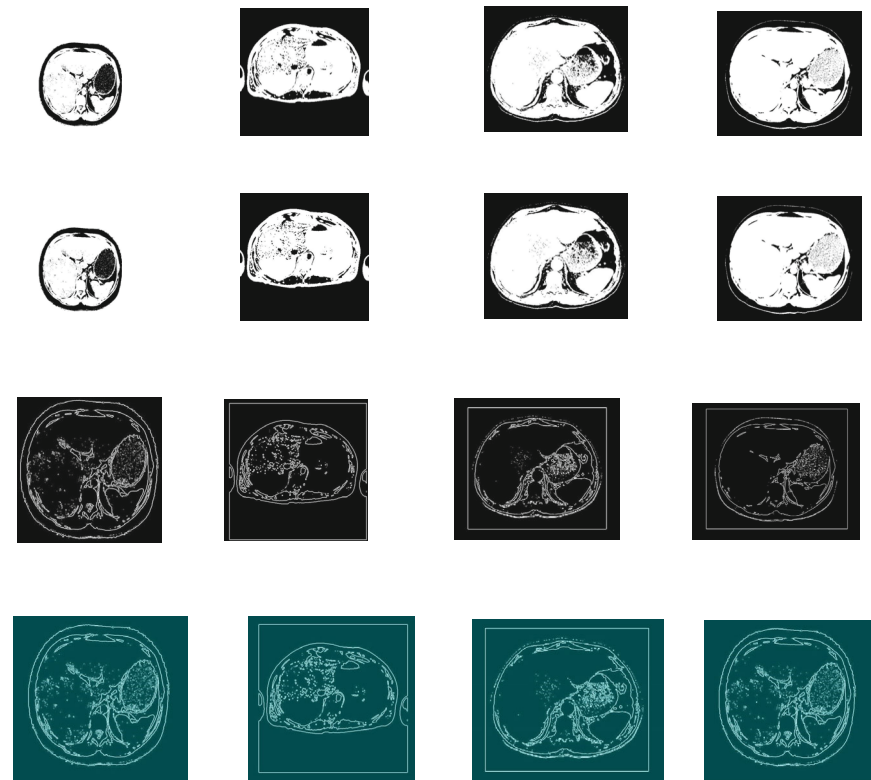


Fig. 4 Results of proposed methodology after reading the original images in stepwise

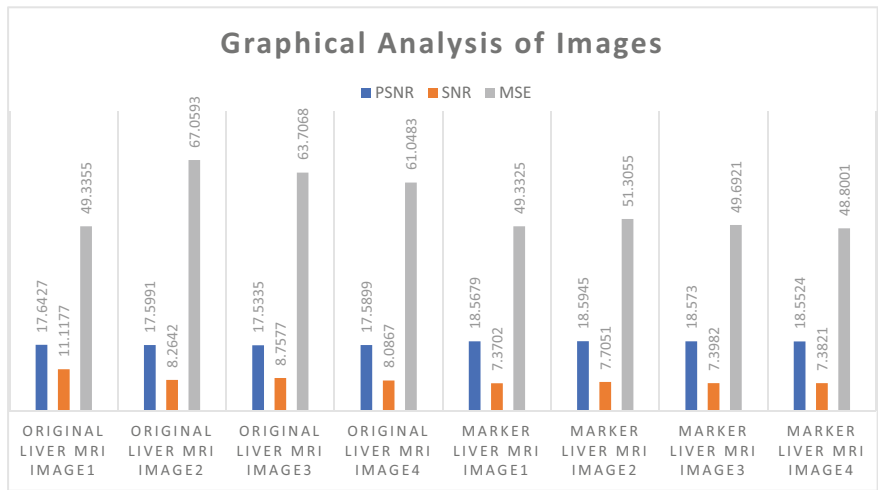


Fig. 5 Graphical analysis of proposed algorithm results

Table 1 Statistical measurement table of proposed algorithm results

Images	PSNR	SNR	MSE
Original liver MRI Image1	17.6427	11.1177	49.3355
Original liver MRI Image2	17.5991	8.2642	67.0593
Original liver MRI Image3	17.5335	8.7577	63.7068
Original liver MRI Image4	17.5899	8.0867	61.0483
Marker liver MRI Image1	18.5679	7.3702	49.3325
Marker liver MRI Image2	18.5945	7.7051	51.3055
Marker liver MRI Image3	18.5730	7.3982	49.6921
Marker liver MRI Image4	18.5524	7.3821	48.8001

Through the statistical table and graphical representation, we can easily see how the algorithm has been worked. We have taken three statistical aspects to measure the level of algorithm working efficiency that are PSNR, SNR, MSE for two different type of Liver MRI images with the dimension of 900×772 and 225×225 . All the resultant values of two different type of MRI images shown that the algorithm can work of different aspect of MRI image if we want to work. All the values have improved after applying the proposed algorithm as we can see in the above table. Also, the original and marker resultant images histogram shown in following to describe the analysis in a more detailed and enhanced way for the effected segmentation approach which will help in identifying the image properties with their brightness. The histogram mainly helps to show the graphs where the number of pixels in an image at each different intensity value can found in that image which will also can get in resultant image after algorithm has used. Figures 6 and 7 are showing in the following actually are the original intensity values of images through histogram analysis and showing the resultant image intensity values which can help to know the reduction process of noises.

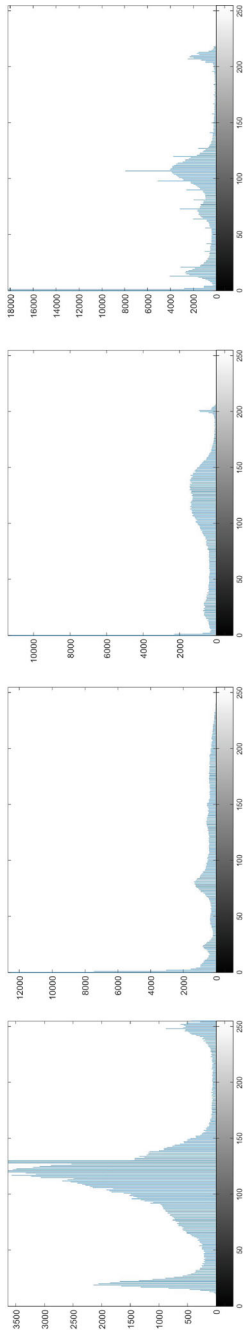


Fig. 6 Original image histograms

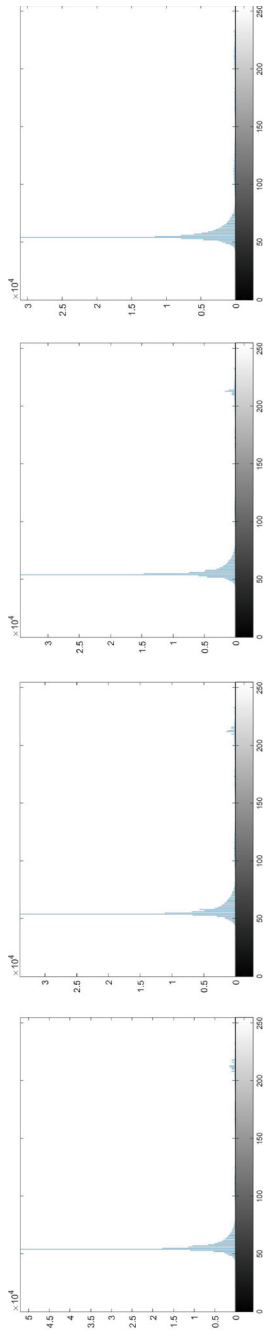


Fig. 7 Marker image histograms

5 Conclusion

After all the experimental works on different types on Liver MRI images we can say we are able to segment the images properly by reducing the noise from any complicated image which normally not possible to segment in a conventional way which leads to early disease detection. The proposed method also enhancing the part of edge detection very well so that the marker can work properly and mark all the working regions. If we are able to reduce the over-segmentation process, we can also reduce the noise from images and able to improve the quality of image by producing more quality pixels. The alignments of markers are very important to work on regions with the help of edge detection. We can offer Sobel approach for MRI image analysis for the difficult image analysis process and do the early diagnosis of any medical diagnosis.

6 Future Scope

In future work the research work may explore the use of deep learning-based denoising methods to effectively handle complex imaging scenarios and develop automated systems for precise marker alignment using advanced feature matching techniques to ensure accurate ROI identification in complex images.

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